Living Arrangements and Wellbeing Analytics Portfolio Project

Dataset Source: https://www.kaggle.com/datasets/asmasohail/integrated-lifestyle-sleep-mental-health

Tableau Public Interactive Dashboard:
https://public.tableau.com/app/profile/larson.st.cyr/viz/LivingArrangementsandWellbeing/LivingArrangementsandWellbeingADataAnalysis

Feel free to explore the interactive dashboard above and see how living arrangements influence wellbeing.

As a Data Analytics professional, I often approach everyday questions through an analytical lens. Recently, a recurring question in my personal life sparked curiosity and inspired this project: *"Can we move in together yet?"* While many sources advise couples not to live together before marriage, I wanted to explore what the data shows. Instead of relying on opinions, I analyzed a wellbeing dataset to examine whether different living arrangements such as cohabiting with a partner, living separately, or living with roommates have any measurable relationship with self-reported happiness and overall wellbeing.

1. Analytical Question

How does living arrangement (cohabiting with a partner, living apart from a partner, or living with roommates) relate to self-reported happiness and wellbeing?

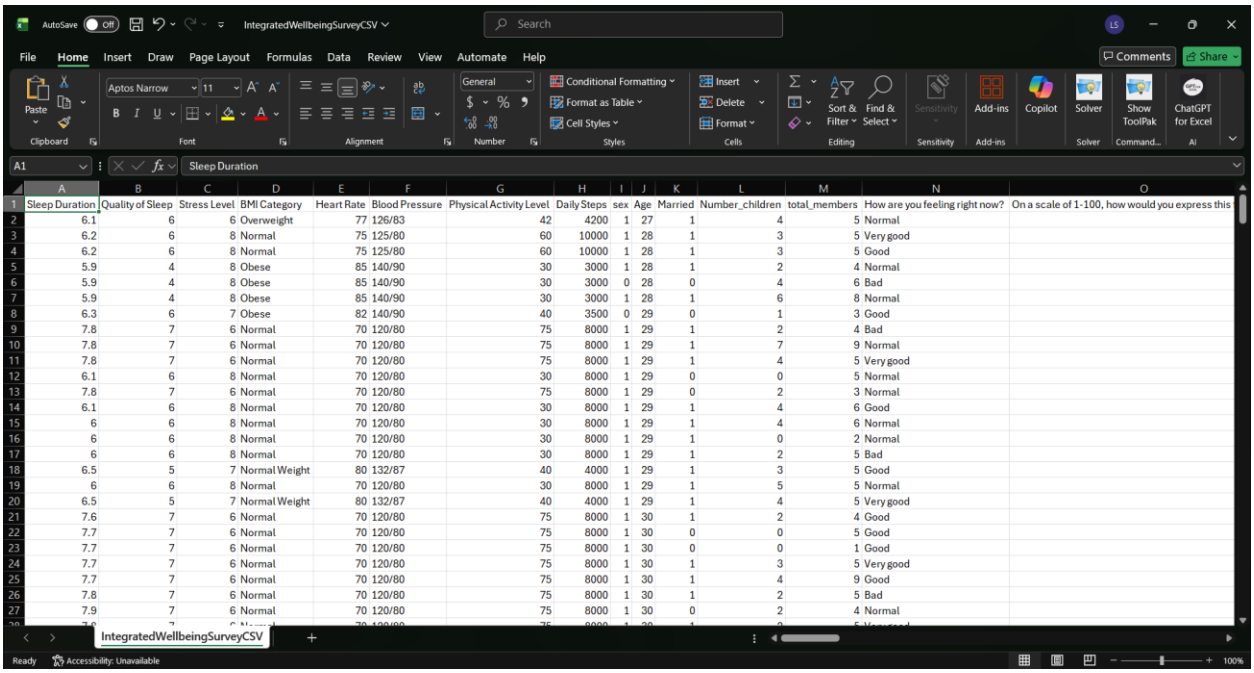
2. Dataset Source

The dataset used for this analysis was obtained from Kaggle and is titled: Integrated Lifestyle, Sleep, and Mental Health Dataset.

This dataset is a merged combination of two separate datasets that capture various lifestyle, physiological, and psychological indicators related to personal wellbeing.

3. Dataset Information
    - Total Records: 375 rows
    - Total Variables: 44 columns
    - Total Exact Duplicates: 16 rows
    - Data Types: Mixed numerical and categorical variables
    - Categories Represented:
        i. Psychological wellbeing indicators
        ii. Lifestyle and health behaviors
        iii. Sleep patterns and disorders
        iv. Demographic and social variables

This data set provides a comprehensive set of variables that enable analysis of how living arrangements relate to self-reported happiness, stress levels, sleep quality, and overall wellbeing.



Sleep Duration	Quality of Sleep	Stress Level	BMI Category	Heart Rate	Blood Pressure	Physical Activity Level	Daily Steps	sex	Age	Married	Number_children	total_members	How are you feeling right now?	On a scale of 1-100, how would you express this
6.1	6	6	Overweight	77	126/83	42	4200	1	27	1	4	5	Normal	
6.2	6	8	Normal	75	125/80	60	10000	1	28	1	3	5	Very good	
6.2	6	8	Normal	75	125/80	60	10000	1	28	1	3	5	Good	
5.9	4	8	Obese	85	140/90	30	3000	1	28	1	2	4	Normal	
5.9	4	8	Obese	85	140/90	30	3000	0	28	0	4	6	Bad	
5.9	4	8	Obese	85	140/90	30	3000	1	28	1	6	8	Normal	
6.3	6	7	Obese	82	140/90	40	3500	0	29	0	1	3	Good	
7.8	7	6	Normal	70	120/80	75	8000	1	29	1	2	4	Bad	
7.8	7	6	Normal	70	120/80	75	8000	1	29	1	7	9	Normal	
7.8	7	6	Normal	70	120/80	75	8000	1	29	1	4	5	Very good	
6.1	6	8	Normal	70	120/80	30	8000	1	29	0	0	5	Normal	
7.8	7	6	Normal	70	120/80	75	8000	1	29	0	2	3	Normal	
6.1	6	8	Normal	70	120/80	30	8000	1	29	1	4	6	Good	
6	6	8	Normal	70	120/80	30	8000	1	29	1	4	6	Normal	
6	6	8	Normal	70	120/80	30	8000	1	29	1	0	2	Normal	
6	6	8	Normal	70	120/80	30	8000	1	29	1	2	5	Bad	
6.5	5	7	Normal Weight	80	132/87	40	4000	1	29	1	3	5	Good	
6	6	8	Normal	70	120/80	30	8000	1	29	1	5	5	Normal	
6.5	5	7	Normal Weight	80	132/87	40	4000	1	29	1	4	5	Very good	
7.6	7	6	Normal	70	120/80	75	8000	1	30	1	2	4	Good	
7.7	7	6	Normal	70	120/80	75	8000	1	30	0	0	5	Good	
7.7	7	6	Normal	70	120/80	75	8000	1	30	0	0	1	Good	
7.7	7	6	Normal	70	120/80	75	8000	1	30	1	3	5	Very good	
7.7	7	6	Normal	70	120/80	75	8000	1	30	1	4	9	Good	
7.8	7	6	Normal	70	120/80	75	8000	1	30	1	2	5	Bad	
7.9	7	6	Normal	70	120/80	75	8000	1	30	0	2	4	Normal	

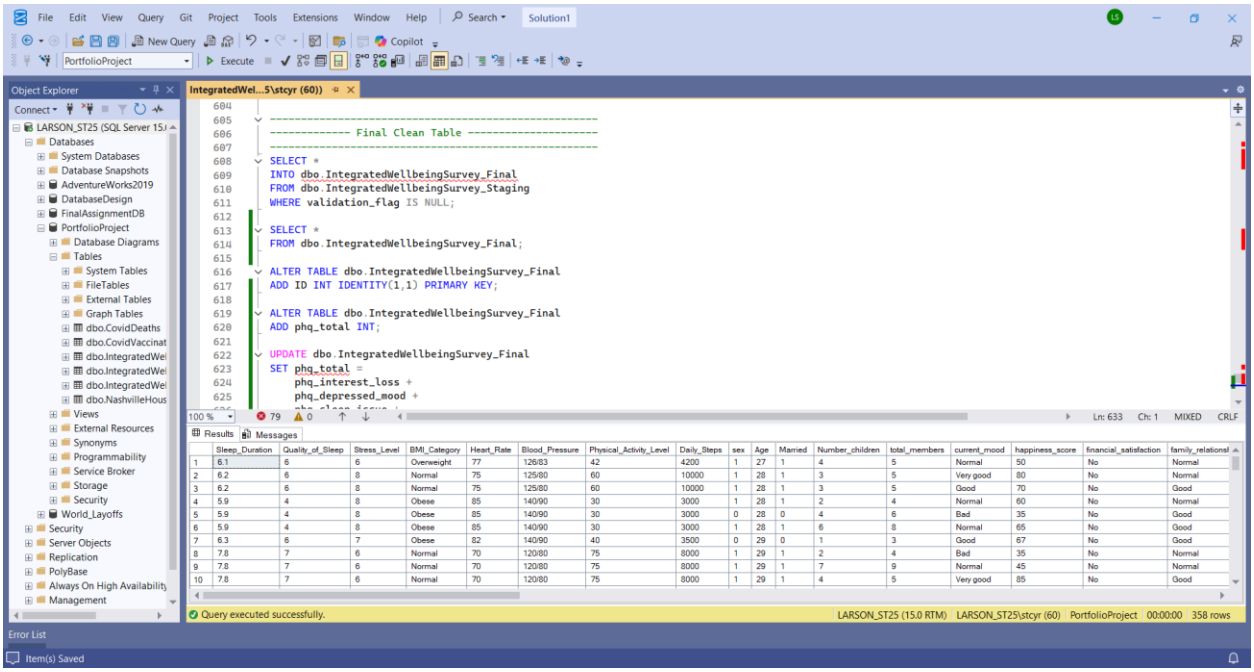
4. Data Cleaning Approach

The Integrated Wellbeing Survey dataset contained many survey variables and responses, which required a structured approach to transform the raw data into an analysis-ready dataset. With 44 columns of information, it was important to remain focused on the objective of preparing reliable and consistent data for analysis rather than becoming overwhelmed by the volume of available variables.

- To perform the cleaning process, I used **Microsoft SQL Server Management Studio (SSMS)** to stage, audit, and standardize the dataset. SQL allowed me to efficiently inspect data quality issues and apply systematic transformations across the entire dataset.
- Key cleaning tasks included correcting inconsistent data types, handling missing values, validating category consistency, and identifying duplicate records. A staging table was created so the original dataset remained unchanged while cleaning operations were performed safely.
- Duplicate auditing revealed several identical records, which were reviewed and removed to ensure the dataset contained only unique observations.
- Certain categorical fields also require standardization. For example, the **BMI Category** column contained both *"Normal"* and *"Normal Weight."* These values were standardized to *"Normal"* to maintain consistency during grouping and analysis.
- Blank entries and placeholder values such as *"N/A," "NA,"* and *"None"* were converted to **NULL values** to prevent inaccurate aggregations and artificial categories in later analysis.
- Because the dataset consisted of many survey-based questions with lengthy column names, several columns were renamed to shorter, analysis-friendly variable names. This

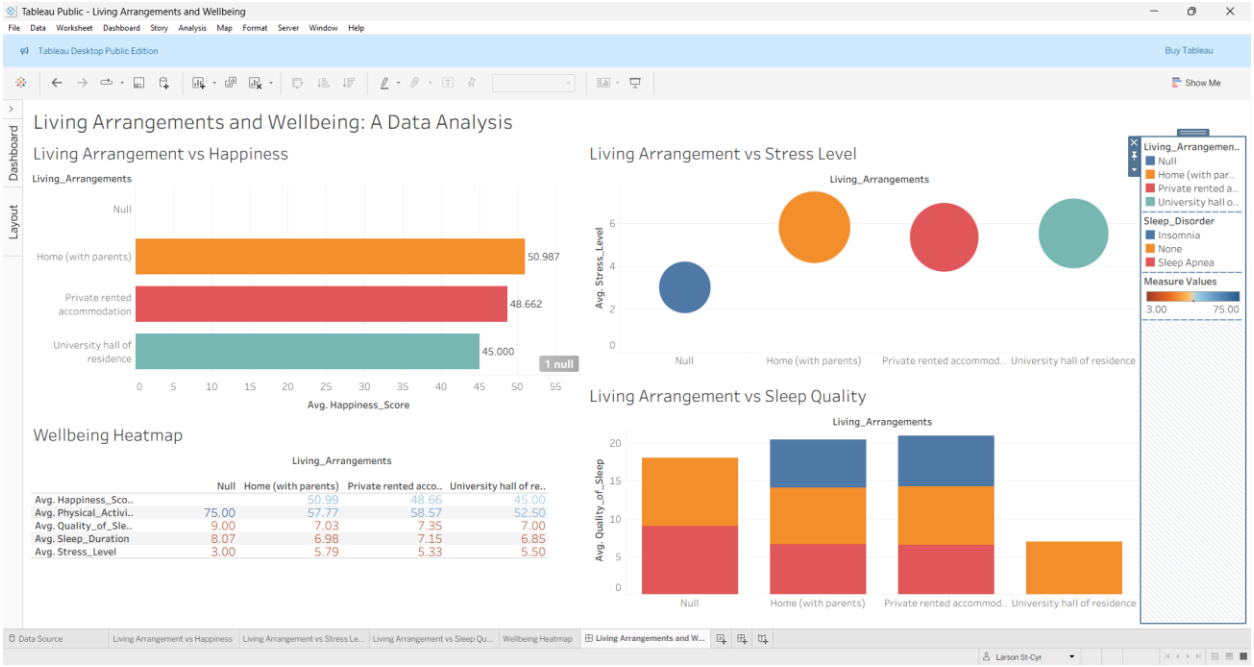
improved readability, simplified SQL queries, and aligned the dataset with common analytical naming conventions.

Through this cleaning process, the dataset was transformed from raw survey data into a structured dataset suitable for analysis and visualization in Tableau.



5. Key Insights from Dashboard
   - After analyzing the dataset in Tableau, I noticed these patterns:
     i. Individuals cohabiting at Home (with their parents) generally reported higher happiness scores compared to those that Privately Rented or at the University Hall.
     ii. It's interesting to note that people who live at home (with their parents) exhibit somewhat higher stress levels than others since they sleep less and with lower quality.
     iii. Individuals who have privately rented accommodation have the highest physical activity levels (58.6) compared to those living at home (57.8) or in university housing (52.5)

This interactive Tableau dashboard analyzes how living arrangements influence self-reported wellbeing indicators such as happiness, stress level, and sleep quality. Users can explore the data through filters including sex, marital status, and depression indicators to examine how demographic factors influence wellbeing across different living arrangements.



Summary:

This project involved performing full end-to-end data cleaning, transformation, and validation on a lifestyle and depression dataset using SQL Server.

The process included:

- Data cleaning
- Validation logic
- Data Engineering
- Database design
- BI visualization
- Analytical storytelling
- Creation of a final analytics-ready SQL table

The cleaned dataset was then integrated into Tableau to develop visualizations exploring relationships between sleep, stress, physical activity, and depression outcomes.

This project demonstrates proficiency in SQL-based ETL workflows, data validation, feature engineering, and business intelligence visualization.